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Tutorial 10

1) $L = \{0^n 1^n \mid n \geq 1\}$ $Q = \{q_0, q_1, q_T, q_A\}$

$\delta(q_0, \varepsilon, \varepsilon) \to q_1, \$$

$\delta(q_1, 0, \$) \to q_1, 0\$$

$\delta(q_1, 0, 0) \to q_1, 00$

$\delta(q_1, 0, 1) \to q_T, \varepsilon$

$\delta(q_1, 1, \$) \to q_T, \varepsilon$

$\delta(q_1, 1, 0) \to q_1, \varepsilon$

$\delta(q_1, 1, 1) \to q_T, \varepsilon$

$\delta(q_T, 0|1|\varepsilon, *) \to q_T, \varepsilon$

$\delta(q_1, \varepsilon, \$) \to q_A, \varepsilon$

2) $L = \{a^n b^n \mid n \geq 0\}$

When the first letter is a

$\delta(q_0, \varepsilon, \varepsilon) \to q_1, \$$

$\delta(q_1, a, \$) \to q_1, a\$$

$\delta(q_1, a, a) \to q_1, aa$

$\delta(q_1, a, b) \to q_T, \varepsilon$

$\delta(q_1, b, \$) \to q_T, \varepsilon$

$\delta(q_1, b, a) \to q_1, \varepsilon$

$\delta(q_1, b, b) \to q_T, \varepsilon$

When the first letter is b

$\delta(q_0, \varepsilon, \varepsilon) \to q_1, \$$

$\delta(q_1, b, \$) \to q_1, b\$$

$\delta(q_1, b, b) \to q_1, bb$

$\delta(q_1, b, a) \to q_T, \varepsilon$

$\delta(q_1, a, \$) \to q_T, \varepsilon$

$\delta(q_1, a, b) \to q_1, \varepsilon$

$\delta(q_1, a, a) \to q_T, \varepsilon$

$\delta(q_T, a|b|\varepsilon, *) \to q_T, \varepsilon$

$\delta(q_1, \varepsilon, \$) \to q_A, \varepsilon$ //

3) $L = \{ww^R \mid w = (a+b)^+\}$

$\delta(q_0, \varepsilon, \varepsilon) \to q_1, \$$

$\delta(q_1, a, \$) \to q_1, a\$$

$\delta(q_1, a, a) \to q_1, \varepsilon$

$\delta(q_1, a, b) \to q_1, ab$

$\delta(q_1, b, \$) \to q_1, b\$$

$\delta(q_1, b, b) \to q_1, \varepsilon$

$\delta(q_1, b, a) \to q_1, ba$

$\delta(q_1, \varepsilon, \$) \to q_A, \varepsilon$ //

4) $L = \{wcw^R\}$ $\Sigma = \{a, b, c\}$ assumption: $w \neq \varepsilon$

$Q = \{q_0, q_1, q_2, q_T, q_A\}$

$\delta(q_0, \varepsilon, \varepsilon) \to q_1, \$$

$\delta(q_1, a, \$) \to q_1, a\$$

$\delta(q_1, b, \$) \to q_1, b\$$

$\delta(q_1, a, a) \to q_1, aa$

$\delta(q_1, a, b) \to q_1, ab$

$\delta(q_1, b, a) \to q_1, ba$

$\delta(q_1, b, b) \to q_1, bb$

$\delta(q_1, c, a) \to q_2, \varepsilon$

$\delta(q_1, c, b) \to q_2, \varepsilon$

$\delta(q_1, c, \$) \to q_T, \varepsilon$

$\delta(q_2, a, a) \to q_2, \varepsilon$

$\delta(q_2, a, b) \to q_T, \varepsilon$

$\delta(q_2, b, b) \to q_2, \varepsilon$

$\delta(q_2, b, a) \to q_T, \varepsilon$

$\delta(q_2, b, \$) \to q_T, \varepsilon$

$\delta(q_2, a, \$) \to q_T, \varepsilon$

$\delta(q_2, \varepsilon, \$) \to q_A, \varepsilon$

$\delta(q_T, a|b|\varepsilon, *) \to q_T, \varepsilon$ //

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5) $L = a^m b^n a^m \quad n, m > 1$

$Q = (q_0, q_1, q_2, q_3, q_T, q_A)$

$\delta(q_0, \varepsilon, \varepsilon) \to q_1, \$$
$\delta(q_1, a, \$) \to q_1, a\$$
$\delta(q_1, b, \$) \to q_1, b\$$
$\delta(q_1, a, a) \to q_1, aa$
$\delta(q_1, a, b) \to q_T, \varepsilon$
$\delta(q_1, b, a) \to q_2, \varepsilon$
$\delta(q_1, b, b) \to q_T, \varepsilon$
$\delta(q_2, b, b) \to q_T, \varepsilon$
$\delta(q_2, b, a) \to q_2, \varepsilon$
$\delta(q_2, b, \$) q_T, \varepsilon$
$\delta(q_2, a, \$) \to q_T, \varepsilon$
$\delta(q_2, a, b) \to q_T, \varepsilon$
$\delta(q_3, a, a) \to q_3, \varepsilon$
$\delta(q_3, a, \$) \to q_T, \varepsilon$
$\delta(q_3, b, *) \to q_T, \varepsilon$
$\delta(q_2, a, a) \to q_3, \varepsilon$
$\delta(q_3, a, b) \to q_T, \varepsilon$
$\delta(q_3, \varepsilon, \$) \to q_A, \varepsilon$

6) $L = a^{m+n} b^m c^n$

$\Sigma = (a, b, c) \quad m,n > 1$

$Q = (q_0, q_1, q_2, q_3, q_T, q_A)$

$\delta(q_0, \varepsilon, \varepsilon) \to q_1, \$$
$\delta(q_1, a, \$) \to q_1, a\$$
$\delta(q_1, b, \$) \to q_T, \varepsilon$
$\delta(q_1, c, \$) \to q_T, \varepsilon$
$\delta(q_1, a, a) \to q_1, aa$
$\delta(q_1, a, b|c) \to q_T, \varepsilon$
$\delta(q_1, b, a) \to q_2, \varepsilon$
$\delta(q_2, c, a|c) \to q_T, \varepsilon$
$\delta(q_2, a, *) \to q_T, \varepsilon$
$\delta(q_2, b, a) \to q_2, \varepsilon$
$\delta(q_2, c, a) \to q_3, \varepsilon$
$\delta(q_2, b, c|b) \to q_T, \varepsilon$
$\delta(q_3, a|b, *) \to q_T, \varepsilon$
$\delta(q_3, c, a) \to q_3, \varepsilon$
$\delta(q_3, c, b|c) \to q_T, \varepsilon$
$\delta(q_3, \varepsilon, \$) \to q_A, \varepsilon$
$\delta(q_T, a|b|c|\varepsilon, *) \to q_T, \varepsilon$

7) $L = a^n b^{2n} \quad n \geq 1$

$\Sigma = (a, b)$

$Q = (q_0, q_1, q_2, q_A, q_T)$

$\delta(q_0, \varepsilon, \varepsilon) \to q_1, \$$
$\delta(q_1, a, \$) \to q_1, a\$$
$\delta(q_1, b, \$) \to q_T, \cancel{ba}\ \varepsilon$
$\delta(q_1, a, a) \to q_1, aa$
$\delta(q_1, a, b) \to q_T, \varepsilon$
$\delta(q_1, b, b|\$) \to q_T, \varepsilon$
$\delta(q_1, b, a) \to q_2, ba$
$\delta(q_2, b, b) \to q_2, \varepsilon$
$\delta(q_2, b, a) \to q_2, ba$
$\delta(q_2, a, *) \to q_T, \varepsilon$
$\delta(q_2, \varepsilon, \$) \to q_A, \varepsilon$
$\delta(q_T, *, *) \to q_T, \varepsilon$

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8) $L = a^m b^n c^{m+n}$ $\Sigma = (a, b, c)$

$m, n \geq 1$

$Q = (q_0, q_1, q_2, q_3, q_T, q_A)$

$\delta(q_0, \varepsilon, \varepsilon) \rightarrow q_1, \$$

$\delta(q_1, a, \$) \rightarrow q_1, a\$$

$\delta(q_1, b|c, \$) \rightarrow q_T, \varepsilon$

$\delta(q_1, a, a) \rightarrow q_1, aa$

$\delta(q_1, a, b|c) \rightarrow q_T, \varepsilon$

$\delta(q_1, b, a) \rightarrow q_2, ba$

$\delta(q_1, c, *) \rightarrow q_T, \varepsilon$

$\delta(q_1, b, b|c) \rightarrow q_T, \varepsilon$

$\delta(q_2, b, b) \rightarrow q_2, bb$

$\delta(q_2, a, *) \rightarrow q_T, \varepsilon$

$\delta(q_2, b, a|c|\$) \rightarrow q_T, \varepsilon$

$\delta(q_2, c, b) \rightarrow q_3, \varepsilon$

$\delta(q_2, c, a|\$) \rightarrow q_T, \varepsilon$

$\delta(q_3, c, a) \rightarrow q_3, \varepsilon$

$\delta(q_3, c, b) \rightarrow q_3, \varepsilon$

$\delta(q_3, c, c|\$) \rightarrow q_T, \varepsilon$

$\delta(q_3, \varepsilon, \$) \rightarrow q_A, \varepsilon$

$\delta(q_T, \varepsilon^*, *) \rightarrow q_T, \varepsilon$

9) $L = a^n, b^m, c^m$ $\Sigma = (a, b, c)$

$n, m \geq 1$

$\delta(q_0, \varepsilon, \varepsilon) \rightarrow q_1, \$$

$\delta(q_1, a, \$) \rightarrow q_1, a\$$

$\delta(q_1, a, a) \rightarrow q_1, aa$

$\delta(q_1, a, b|c) \rightarrow q_T, \varepsilon$

$\delta(q_1, c, *) \rightarrow q_T, \varepsilon$

$\delta(q_1, b, a) \rightarrow q_2, ba$

$\delta(q_1, b, b|c|\$) \rightarrow q_T, \varepsilon$

$\delta(q_2, b, b) \rightarrow q_2, bb$

$\delta(q_2, a, *) \rightarrow q_T, \varepsilon$

$\delta(q_2, c, a|c|\$) \rightarrow q_T, \varepsilon$

$\delta(q_2, c, b) \rightarrow q_3, \varepsilon$

$\delta(q_3, \cancel{b}, a) \rightarrow q_T, \varepsilon$

$\delta(q_3, a|b, *) \rightarrow q_T, \varepsilon$

$\delta(q_3, \cancel{c}\ \varepsilon, a) \rightarrow q_A, \varepsilon$

$\delta(q_T, *, *) \rightarrow q_T, \varepsilon$

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